Support Vector Machines for Image Classification on the CIFAR-10 Dataset

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***Abstract*—In this project we study Support Vector Machines (SVMs) on the CIFAR-10 image dataset. Two classification tasks are examined:**

1) **Binary classification: separation of 2 (out of the 10 in total) CIFAR-10 classes. For our experiments, we chose classes 3 and 5 (cat vs dog).**
2) **Multiclass classification: separation of all 10 CIFAR-10 classes.**

Several SVM kernels (linear / polynomial / radial basis function (RBF) / sigmoid) are examined, as well as different hyperparameters. Also, we compare SVM performance against three baselines:

1) **Nearest Class Centroid**
2) **k-Nearest Neighbors for k=1,3**
3) **MLP with 1 hidden layer trained with a hinge loss**

I Introduction

In this report, we present our findings on the application of Support Vector Machines (SVMs) for binary and multiclass image classification tasks using the CIFAR-10 dataset. The goal of this project is to evaluate and analyze how different SVM kernels perform on high-dimensional data and understand the impact of the various hyperparameters, such as the regularization parameter C and kernel-specific parameters like γ for the RBF kernel and degree d for the polynomial kernel. We also compare the performance of SVMs against several baseline classifiers, including Nearest Centroid, k-Nearest Neighbors (k-NN), and a shallow Multi-Layer Perceptron (MLP) trained with hinge loss.

II Dataset

A. CIFAR-10

CIFAR-10 contains 60,000 RGB images of size 32x32 (so 3x32x32) belonging to 10 classes: airplane, automobile, bird, cat, deer, fog, frog, horse, ship, truck. The dataset split is 50,000 samples for training and 10,000 for testing.

B. Data Preproccesing

- Flattening
  Each image, as mentioned before, is a 3x32x32 tensor. We flatten the images into vectors of length 3072
- Standardization
  We apply StandardScaler to the flattened features so each feature has zero mean and unit variance. This step is important for the SVMs, especially with RBF and polynomial kernels and for distance-based methods(k-NN, Nearest Centroid).

C. Dimensionality Reduction with PCA

Due to the high dimensionality of the raw images, we apply Principal Component Analysis to retain 90% of the dataset variance. The number of components retained is 103 out of the 3072 in total. Since SVMs can be computationally expensive with high-dimensional data, this step helps reduce computational costs and potentially improve model performance by removing noise and redundant features. Since PCA is a linear transformation, it preserves the variance rather than the class separability and most certainly it does not guarantee that the classes of the CIFAR-10 dataset will be linearly separable in the reduced space which is the reason we also explore non-linear SVM kernels.

III Support Vector Machines

Support Vector Machines (SVMs) fall into the category of supervised learning algorithms. The main goal of SVMs is to find the best posssible line, or decision boundary, which distinguishes the data samples of different classes. When working on multidimensional data, this boundary is also known as hyperplane and it separates data points belonging to different classes. The main concept of SVMs is to maximize the margin, the distance between the hyperplane and the closest samples of each class, so the dataset is best separated into classes. That is because, in general the larger the margin is, the lower the classifier's generalization error will be. These closest samples to the hyperplane are called Support Vectors.

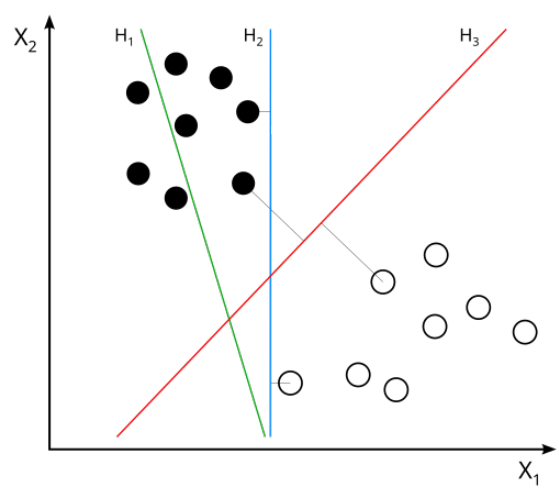


Fig. 1: SVM margin and support vectors

However, in many problems, including ours(as it is later shown from the poor performance of the linear kernel, especially in the multiclass setting), the data is not always

linearly separable. To address and overcome this issue, SVM algorithms apply the so-called **kernel trick**. The kernel trick involves mapping the original low-dimensional data into a higher-dimensional space where they become linearly separable. This transition is achieved through kernel functions, which compute the inner products between the images of all pairs of data points in the feature space, without explicitly performing the transformation in that space. This approach allows SVMs to efficiently handle non-linear decision boundaries. For our experiments, we examine four different kernel functions:

- **Linear Kernel** : $K(x_i, x_j) = x_i^T x_j$
  For linearly separable data. Measures similarity using the dot product and produces more interpretable boundaries.
- **Polynomial Kernel** : $K(x_i, x_j) = (x_i^T x_j + c)^d$ where c is a constant(bias) and d is the degree of the polynomial. Maps the data into a higher-dimensional space by finding the dot products and powers of all features in the dataset.
- **Radial Basis Function (RBF) Kernel** : $K(x_i, x_j) = exp(-\gamma||x_i - x_j||^2)$ where γ is a parameter that defines the how far the influence of a single training example reaches and $||x_i - x_j||^2$ is the squared Euclidean distance between two samples.
  Maps the data into an infinite-dimensional feature and allows for complex decision boundaries.
- **Sigmoid Kernel** : $K(x_i, x_j) = tanh(\alpha x_i^T x_j + c)$ where α is the slope and c is the bias term.
  Behaves like a neural network activation function and shows unstable performance in practice.

IV Experiment setup and pipeline

For our experiments, we used the **SVC** and **LinearSVC** implementation from the **cuML** library, which provides GPU-accelerated algorithms based on scikit-learn implementations. The experiments were conducted in Google Colab using the T4 and L4 GPUs. As mentioned earlier, we explored both binary and multiclass classification tasks on the CIFAR-10 dataset. For each task we evaluated the performance of different SVM kernels and hyperparameters. Also, for the multiclass classification task, we used both one-vs-one (OvO) and one-vs-rest (OvR) strategies. Moreover, for each classification task a grid search with 5 (or 3 for computational reasons) -fold cross-validation was performed to find the optimal hyperparameters for the linear and RBF kernels. It is noted that the polynomial and sigmoid kernels were excluded from the grid search due to their high computational costs and time constraints plus the fact that they generally perform worse than the RBF kernel. The examined hyperparameters were the regularization parameter C, which controls the trade-off between achieving a low training error and a low testing error, and the kernel coefficient γ for the RBF kernel, which defines how far the influence of a single training example reaches. The hyperparameter values evaluated during the grid search were:

- Regularization parameter C: [0.001, 0.01, 0.1, 1, 10, 100, 1000]
- Kernel coefficient γ for RBF kernel: ['scale', 'auto', 0.001, 0.01, 0.1, 1]
  where 'scale' uses $1/(n_{features} * X.\text{variance}())$ as value of γ and 'auto' uses $1/n_{features}$, which in our case $n_{features} = 103$ after PCA.

Also, we evaluated the performance of the polynomial kernel for degrees $d = [2, 3, 4, 5]$. For sigmoid kernel, we used the default hyperparameters provided by cuML. Finally, we compared the SVM performance against three baselines:

- Nearest Class Centroid
- k-Nearest Neighbors for k=1,3
- MLP with 1 hidden layer of 128 neurons trained with a hinge loss for 50 epochs
  Hinge loss, which is also the default loss function for SVMs, is defined as:

$$L(y, f(x)) = max(0, 1 - y \cdot f(x)) \tag{1}$$

  where y is the true label and $f(x)$ is the predicted output. It should be clarified thath the MLP architecture used is quite simple and shallow intentionally, as the goal is to provide a baseline for comparison rather than exploring more complex neural network architectures (such as deeper MLPs or CNNs as we encountered in the previous project) which would most likely outperform SVMs.

The overall experiment pipeline is the following:

1) Load CIFAR-10 dataset to GPU memory
2) Preprocess data: flattening, standardization, PCA
3) Create binary subset for classes 3 and 5 (cat vs dog)
4) For each classification task (binary / multiclass):
   a) For each SVM kernel (linear / polynomial / RBF / sigmoid):
      i) Train SVM model on the training set with hyperparameters as stated above and evaluate on the test set (42 different models in total for binary classification and 84 for multiclass)
      ii) If multiclass classification, try both OvO and OvR strategies
      iii) If kernel is linear or RBF:
         A) Perform grid search with cross-validation to find optimal hyperparameters
         B) Train SVM model with optimal hyperparameters on the training set
         C) Evaluate model on the test set and record accuracy
      iv) For each baseline classifier (Nearest Centroid / 1-NN, 3-NN / MLP):
         A) Train model on the training set
         B) Evaluate model on the test set and record accuracy
   b) Compare SVM results against baselines

Each experiment's recorded metric is the classification accuracy on the test set. Also, training time is recorded for each model to compare computational costs. For the grid search experiments, the best hyperparameters found are recorded for each problem type. Every model and its results are stored for future reference.

V Results

In this section, we present the results of our experiments on both binary and multiclass classification tasks using Support

Vector Machines (SVMs) with different kernels, as well as the performance of baseline classifiers.

A. Binary Classification Results

For the binary classification task we focused on distinguishing between the 'cat' and 'dog' classes (classes 3 and 5) from the CIFAR-10 dataset. The top results per kernel are summarized in Table I. The full results for all examined SVM models are presented in Table VI in the Appendix.

Model	Min	Sec	Train Acc.	Test Acc.
RBF(γ = scale, $C = 10$)	0	0	0.9681	0.6740
Poly(degree = 2, $C = 10$, γ = scale)	0	0	0.8079	0.6530
Poly(degree = 3, $C = 1$, γ = scale)	0	0	0.8156	0.6470
Poly(degree = 5, $C = 10$, γ = scale)	0	0	0.9702	0.6255
Poly(degree = 4, $C = 10$, γ = scale)	0	0	0.9723	0.6215
Linear($C = 100$)	0	0	0.6281	0.6100
Sigmoid(γ = scale, $C = 0.01$)	0	0	0.5960	0.5905

TABLE I: Performance comparison of best SVM models per kernel for binary classification (cat vs dog)

For the binary classification task, we observed varying performance across different SVM kernels and hyperparameter settings. We have to state that the results reported here are based on the test accuracy achieved on the test set after training the models on the training set. As we can see from Table I, the RBF kernel outperforms the other kernels in terms of test accuracy, achieving a test accuracy of 67.40% with $C = 10$ and γ = scale. Here, we have to address that the γ parameter set to 'scale' is equal to 0.00033753205207176507 given that after PCA we have 103 features. We opted for the 'scale' value as it generally provides better performance compared to 'auto' and fixed values in our experiments, because it adapts to the dataset's variance.

The polynomial kernel also shows competitive performance, with the best model achieving a test accuracy of 65.30% for degree 2 and $C = 10$. The linear kernel and sigmoid kernel perform worse in comparison, with test accuracies of 61.00% and 59.05% respectively. Also, it is noted that higher degree polynomial kernels tend to overfit the training data, as indicated by the high training accuracies but lower test accuracies, as seen in degrees 4 and 5. For example, degree 5 with $C = 10$ achieves 97.02% training accuracy but only 62.55% test accuracy (see Table VI in the Appendix for full results). We note that the training times for all SVM models were relatively low, with most models training in under a second, except for a few polynomial kernel models which took slightly longer (due to the higher computational complexity of polynomial kernels).

We also observed that increasing the regularization parameter C generally led to better training accuracies, but did not always translate to improved test accuracies, indicating potential overfitting in some cases. This trend is particularly evident in the RBF kernel, where very high values of C (e.g., 1000) resulted in perfect training accuracy but lower test accuracy (64.80%). The phenomenon can be clearly visualized by plotting the training and test accuracies against different C values for the different kernels. Here, we present such a plot for the RBF kernel in Figure 15 and for the polynomial kernel of degree 2 in Figure 18. For the rest of the kernels, similar plots can be found in the Appendix.

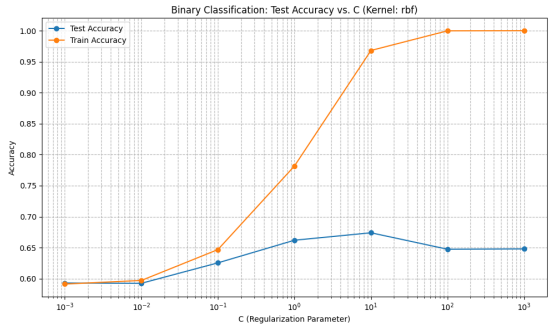


Fig. 2: Binary Classification: RBF Kernel - C vs Accuracy

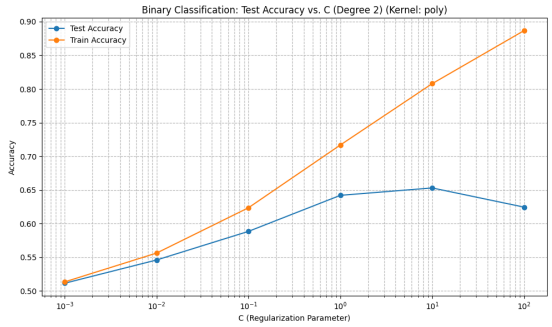


Fig. 3: Binary Classification: Polynomial Kernel (degree=2) - C vs Accuracy

Overall, the RBF kernel with appropriate hyperparameter tuning appears to be the most effective choice for binary classification tasks on the CIFAR-10 dataset. All the results for every examined SVM model for the binary classification task are presented in Table 'VI in the Appendix and a visual comparison of the test accuracies of all models is shown in Figure '22.

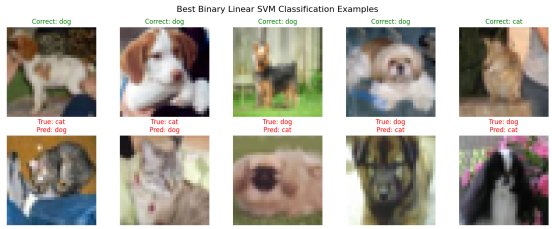


Fig. 4: Binary Classification: Examples of classified images by the best linear SVM model

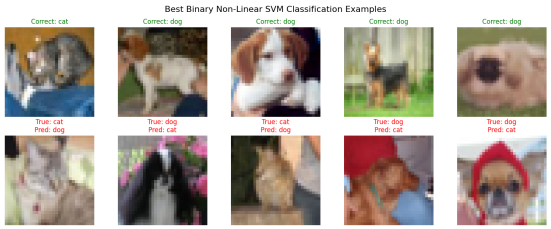


Fig. 5: Binary Classification: Examples of classified images by the best nonlinear (RBF) SVM model

Lastly, we compared the SVM results against the baseline classifiers, with the following accuracies:

Model	Min	Sec	Train Acc.	Test Acc.
Nearest Centroid	0	0	0.5813	0.5840
1-NN	0	0	1.0000	0.5905
3-NN	0	0	0.7967	0.6050
MLP (1 hidden layer, hinge loss)	0	18	0.9783	0.6557

TABLE II: Performance comparison of baseline classifiers for binary classification (cat vs dog)

As we can see from Table II, the MLP classifier with one hidden layer trained with hinge loss achieves the highest test accuracy among the baselines, reaching 65.57%. This performance is competitive with the best SVM models, particularly the polynomial kernel. However, the SVM with RBF kernel still outperforms the MLP, achieving a test accuracy of 67.40%. The k-NN classifiers also perform reasonably well, with 3-NN achieving a test accuracy of 60.50%, but they fall short compared to both the SVM and MLP models. The Nearest Centroid classifier shows the lowest performance among the baselines, with a test accuracy of 58.40%. Overall, the SVM with RBF kernel remains the best-performing model for the binary classification task. In Figure figure 6, we present a visual comparison of the best SVM models against the baseline classifiers in terms of test accuracy.

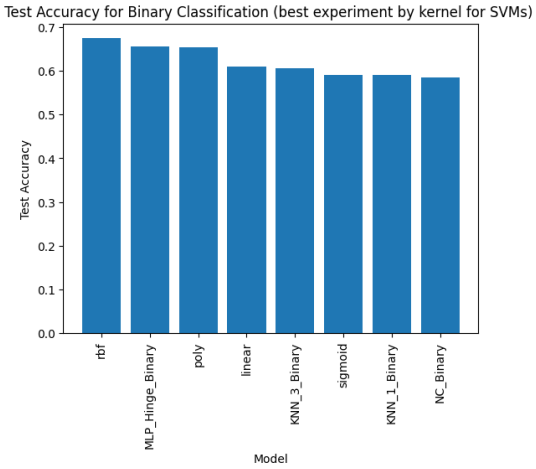


Fig. 6: Binary Classification: SVMs vs Baselines Test Accuracy Comparison

At last, we performed a grid search with 5-fold cross-validation for the linear and RBF kernels to find the optimal hyperparameters. The best hyperparameters found were:

- Linear Kernel: $C = 100$ with test accuracy 60.95%
- RBF Kernel: $C = 1$, $\gamma = 0.001$ with test accuracy 67.70%

The RBF kernel with these optimal hyperparameters slightly outperforms the previously reported best RBF model, indicating the effectiveness of hyperparameter tuning through grid search.

Fig. 7: Binary Classification: Examples of classified images by the MLP baseline model

B. Multiclass Classification Results

For the multiclass classification task, we evaluated the performance of SVMs using both one-vs-one (OvO) and one-vs-rest (OvR) strategies. First of all, it should be noted that the OvO strategy involves training a separate binary classifier for each pair of classes, resulting in a total of $C(10, 2) = 45$ classifiers. In contrast, the OvR strategy trains a single binary classifier for each class, treating that class as the positive class and all other classes as negative, resulting in 10 classifiers.
Regarding the experiments, we examined the same kernels as in the binary classification task. Here we present the top results per kernel and strategy in Table VII and Table VIII respectively. The full results for all examined SVM models and both strategies are presented in the Appendix.

Model	Min	Sec	Train Acc.	Test Acc.
RBF(γ = scale, $C = 10$)	0	9	0.93558	0.5669
Poly($d = 2$, $C = 10$, γ = scale)	0	20	0.70580	0.5024
Poly($d = 3$, $C = 10$, γ = scale)	0	19	0.90086	0.4837
Poly($d = 4$, $C = 100$, γ = scale)	0	18	0.98288	0.4643
Poly($d = 5$, $C = 100$, γ = scale)	0	26	0.96482	0.4400
Linear($C = 10$)	0	0	0.39804	0.3967
Sigmoid(γ = scale, $C = 0.01$)	0	2	0.32220	0.3230

TABLE III: Performance comparison of best SVM models per kernel for multiclass classification (OvO strategy)
Note: γ set to 'scale' is equal to 0.0051441132090985775 given that after PCA we have 103 features.

Model	Min	Sec	Train Acc.	Test Acc.
RBF(γ = scale, $C = 10$)	0	20	0.94534	0.5792
Poly(deg=4, $C = 10$, γ = scale)	1	16	0.96590	0.5135
Poly(deg=5, $C = 10$, γ = scale)	4	45	0.96994	0.5067
Poly(deg=3, $C = 1$, γ = scale)	0	26	0.75720	0.5064
Poly(deg=2, $C = 10$, γ = scale)	0	53	0.66822	0.4969
Linear($C = 0.1$)	0	0	0.39804	0.3967
Sigmoid(γ = scale, $C = 100$)	1	50	0.23834	0.2390

TABLE IV: Performance comparison of best SVM models per kernel for multiclass classification (OvR strategy)
Note: γ set to 'scale' is equal to 0.0051441132090985775 given that after PCA we have 103 features.

From the results presented in Tables III and IV, we observe that the RBF kernel consistently outperforms the other kernels in both OvO and OvR strategies. The best OvO model achieves a test accuracy of 56.69% with $C = 10$ and $\gamma = \text{scale}$, while the best OvR model achieves a slightly higher test accuracy of 57.92% with the same hyperparameters. The polynomial kernel shows moderate performance, with the best OvO model achieving a test accuracy of 50.24% with degree 2 and $C = 10$, and the best OvR model achieving 51.35% with degree 4 and $C = 10$. It is noted that higher degree polynomial kernels tend to overfit the training data, as indicated by the high training accuracies but lower test accuracies. Also, it is interesting to note that the polynomial kernels of lower degrees perform better than those of higher degrees for the One vs One strategy, while the opposite is true for the One vs Rest strategy. The reason behind this phenomenon could be that in the OvR strategy, each classifier has to distinguish one class from all others, which may require more complex decision boundaries that higher-degree polynomials can provide. In contrast, the OvO strategy involves pairwise comparisons between classes, which may be adequately captured by lower-degree polynomials. Finally, the linear and sigmoid kernels perform the worst in both strategies, with test accuracies below 40%. Moreover, as it can be seen in IV in the Appendix, all instances of linear and sigmoid kernels for the OvR strategy performed the worst in comparison with the other kernels with the sigmoid kernel performing the worst overall. Something like this was expected given the complexity of the multiclass classification task and the limitations of these kernels in capturing non-linear decision boundaries. However, it is noted that the linear kernel's performance is consistent across both strategies, achieving a test accuracy of 39.67%. As long for the OvO strategy, the case was not the same with instances of the linear and sigmoid kernel outperforming some weak polynomial kernel models. This could be attributed to the fact that in the OvO strategy, each classifier only needs to distinguish between two classes at a time, which may be more manageable for simpler kernels like linear and sigmoid in certain class pairs. Regarding training times, the OvR strategy generally takes longer to train due to the higher number of classifiers (10 vs 45 in OvO).

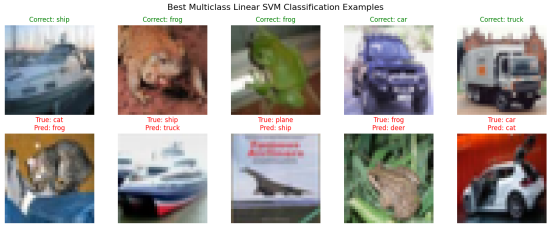


Fig. 8: Multiclass Classification: Examples of classified images by the best linear SVM model

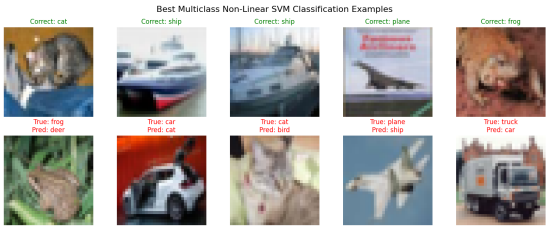


Fig. 9: Multiclass Classification: Examples of classified images by the best nonlinear (RBF) SVM model

We also compared the SVM results against the baseline classifiers for the multiclass classification task, with the following accuracies:

Model	Min	Sec	Train Acc.	Test Acc.
Nearest Centroid	0	0	0.2712	0.2807
1-NN	0	0	100.0	0.3861
3-NN	0	0	0.6209	0.3663
MLP (1 hidden layer, hinge loss)	1	24	0.6121	0.5171

TABLE V: Performance comparison of baseline classifiers for multiclass classification

As we can see from Table V, the MLP classifier with one hidden layer trained with hinge loss achieves the highest test accuracy among the baselines, reaching 51.71%. This performance is competitive with the best SVM models, particularly the polynomial kernel, but falls short compared to the RBF kernel SVMs. The k-NN classifiers also perform reasonably with 1-NN achieving a test accuracy of 38.61% and the Nearest Centroid classifier showing the lowest performance among the baselines, with a test accuracy of 28.07%. Overall, the SVM with RBF kernel remains the best-performing model for the multiclass classification task as well. In Figure figure 10, we present a visual comparison of the best SVM models for each kernel type for both OvO and OvR strategies against the baseline classifiers in terms of test accuracy.

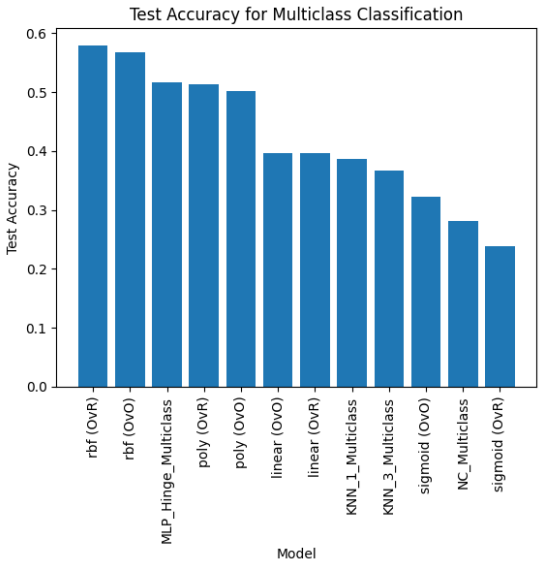


Fig. 10: Multiclass Classification: SVMs vs Baselines Test Accuracy Comparison

Lastly, we performed a grid search with 5-fold cross-validation for the linear and and 3-fold for RBF kernels to find the optimal hyperparameters for the One vs One strategy. The best hyperparameters found were:

- Linear Kernel: $C = 100$ with test accuracy 39.67%
- RBF Kernel: $C = 10$, $\gamma =$ scale with test accuracy 56.7%

The RBF kernel with these optimal hyperparameters matches the previously reported best OvO RBF model, indicating the effectiveness of hyperparameter tuning through grid search. Sadly, due to time constraints, we were not able to perform a grid search for the One vs Rest strategy.

Fig. 11: Multiclass Classification: Examples of classified images by the MLP baseline model

The effects of the C regularization parameter on the training and test accuracies for the multiclass classification task can be visualized by plotting the accuracies against different C values for the different kernels. As we can see in the following figures, the trends are similar to those observed in the binary classification task, with increasing C leading to higher training accuracies but not always improving test accuracies, for both OvO and OvR strategies. For brevity, we present such plots for the RBF kernel in Figures 25 and 32. The plots for the other kernels can be found in the Appendix.

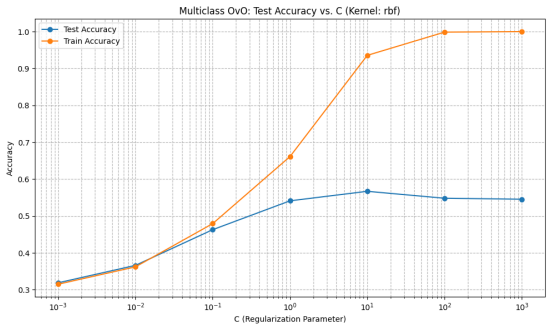


Fig. 12: Multiclass Classification (OvO): RBF Kernel - C vs Accuracy

VI Conclusion

In this report, we explored the application of Support Vector Machines (SVMs) with various kernels on the CIFAR-10 dataset for both binary and multiclass classification tasks. Our experiments demonstrated that the RBF kernel consistently outperformed other kernels, achieving the highest test accuracies in both classification scenarios. The polynomial kernel showed moderate performance but

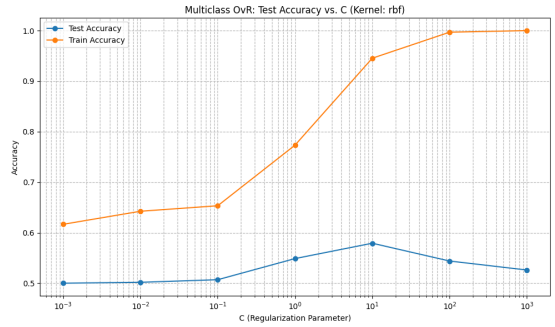
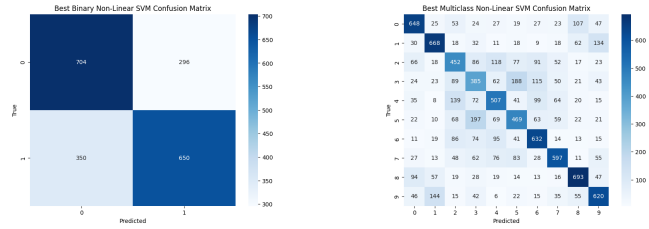


Fig. 13: Multiclass Classification (OvR): RBF Kernel - C vs Accuracy

exhibited proneness to overfitting,while the linear kernel, even though it almost matched the performance of the non-linear kernels in the binary classification task, lagged behind in the multiclass scenario. The sigmoid kernel exhibited the poorest performance across both tasks, while also showing unstable behavior.

Hyperparameter tuning through grid search further enhanced the performance of the RBF kernel, highlighting the importance of selecting appropriate values for parameters such as C and γ.

When compared to baseline classifiers, SVMs with RBF kernel proved to be superior, especially when contrasted with simpler models like Nearest Centroid and k-NN, suggesting that margin-based methods offer significant advantages. The MLP classifier offered competitive performance but still fell short of the best SVM models, although a deeper neural network would probably outperform SVMs.



(a) Binary Classification: Confusion Matrix of the best nonlinear (RBF) SVM model

(b) Multiclass Classification: Confusion Matrix of the best nonlinear (RBF) SVM model

Fig. 14: Confusion matrices for the best nonlinear (RBF) SVM models

Overall, our findings suggest that SVMs, when equipped with appropriate kernels and hyperparameters, can be effective tools for image classification tasks.

Appendix A
Binary Classification Figures and Tables

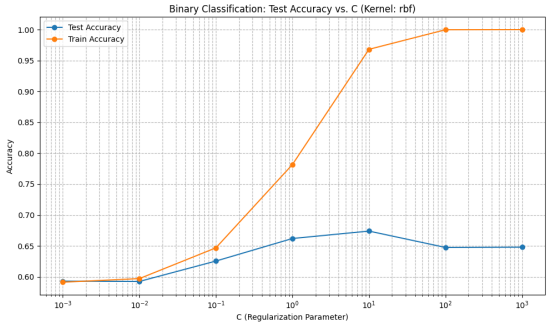


Fig. 15: Binary Classification: RBF Kernel - C vs Accuracy

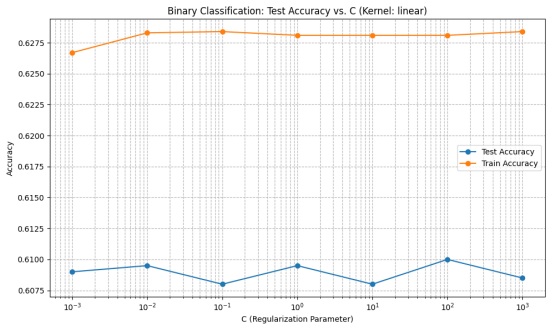


Fig. 16: Binary Classification: Linear Kernel - C vs Accuracy

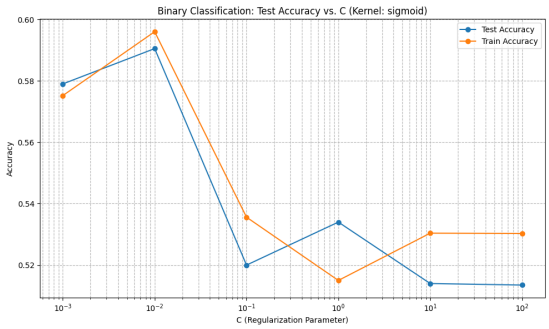


Fig. 17: Binary Classification: Sigmoid Kernel - C vs Accuracy

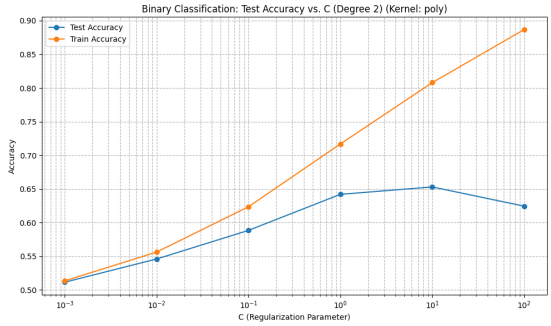


Fig. 18: Binary Classification: Polynomial Kernel (degree=2) - C vs Accuracy

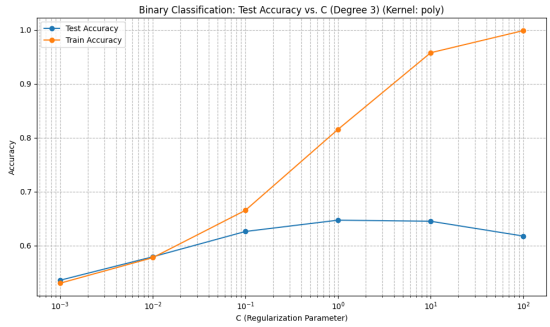


Fig. 19: Binary Classification: Polynomial Kernel (degree=3) - C vs Accuracy

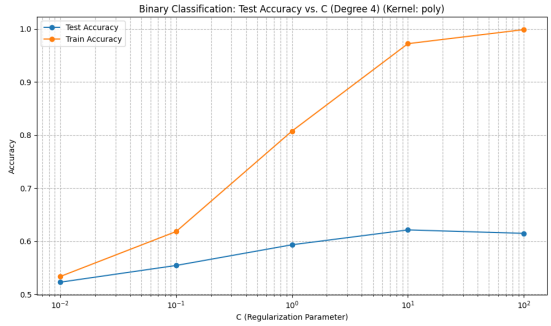


Fig. 20: Binary Classification: Polynomial Kernel (degree=4) - C vs Accuracy

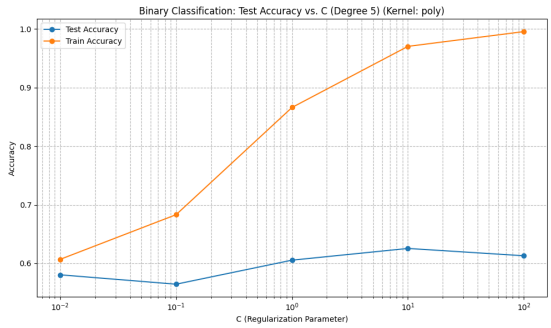


Fig. 21: Binary Classification: Polynomial Kernel (degree=5) - C vs Accuracy

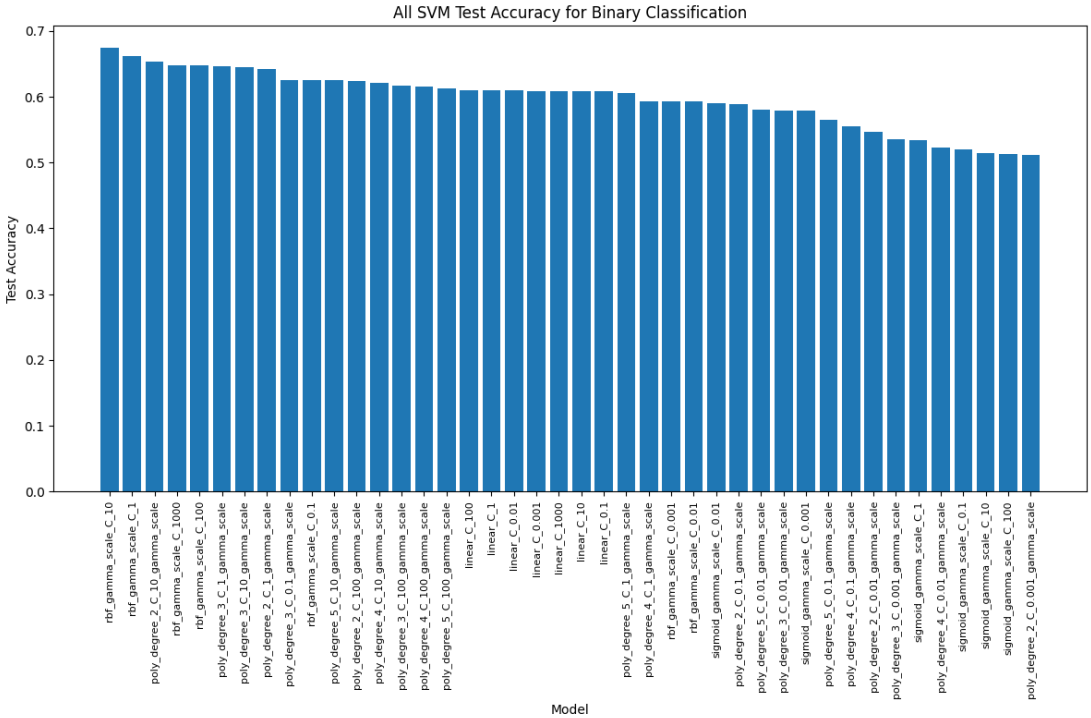


Fig. 22: Binary Classification: All SVM Models Test Accuracy Comparison

Model	Min	Sec	Train Acc.	Test Acc.
RBF(γ = scale, $C = 10$)	0	0	0.9681	0.6740
RBF(γ = scale, $C = 1$)	0	0	0.7817	0.6620
Poly(degree = 2, $C = 10$, γ = scale)	0	0	0.8079	0.6530
RBF(γ = scale, $C = 1000$)	0	0	1.0000	0.6480
RBF(γ = scale, $C = 100$)	0	0	0.9998	0.6475
Poly(degree = 3, $C = 1$, γ = scale)	0	0	0.8156	0.6470
Poly(degree = 3, $C = 10$, γ = scale)	0	0	0.9578	0.6450
Poly(degree = 2, $C = 1$, γ = scale)	0	0	0.7168	0.6420
Poly(degree = 3, $C = 0.1$, γ = scale)	0	0	0.6653	0.6260
Poly(degree = 5, $C = 10$, γ = scale)	0	0	0.9702	0.6255
RBF(γ = scale, $C = 0.1$)	0	0	0.6469	0.6255
Poly(degree = 2, $C = 100$, γ = scale)	0	7	0.8867	0.6245
Poly(degree = 4, $C = 10$, γ = scale)	0	0	0.9723	0.6215
Poly(degree = 3, $C = 100$, γ = scale)	0	2	0.9989	0.6175
Poly(degree = 4, $C = 100$, γ = scale)	0	0	0.9988	0.6150
Poly(degree = 5, $C = 100$, γ = scale)	0	0	0.9954	0.6130
Linear($C = 100$)	0	0	0.6281	0.6100
Linear($C = 0.01$)	0	0	0.6283	0.6095
Linear($C = 1$)	0	0	0.6281	0.6095
Linear($C = 0.001$)	0	0	0.6267	0.6090
Linear($C = 1000$)	0	0	0.6284	0.6085
Linear($C = 0.1$)	0	0	0.6284	0.6080
Linear($C = 10$)	0	0	0.6281	0.6080
Poly(degree = 5, $C = 1$, γ = scale)	0	0	0.8660	0.6055
Poly(degree = 4, $C = 1$, γ = scale)	0	0	0.8077	0.5935
RBF(γ = scale, $C = 0.001$)	0	0	0.5913	0.5930
RBF(γ = scale, $C = 0.01$)	0	0	0.5971	0.5925
Sigmoid(γ = scale, $C = 0.01$)	0	0	0.5960	0.5905
Poly(degree = 2, $C = 0.1$, γ = scale)	0	0	0.6236	0.5885
Poly(degree = 5, $C = 0.01$, γ = scale)	0	0	0.6069	0.5805
Poly(degree = 3, $C = 0.01$, γ = scale)	0	0	0.5774	0.5790
Sigmoid(γ = scale, $C = 0.001$)	0	0	0.5752	0.5790
Poly(degree = 5, $C = 0.1$, γ = scale)	0	0	0.6831	0.5645
Poly(degree = 4, $C = 0.1$, γ = scale)	0	0	0.6184	0.5545
Poly(degree = 2, $C = 0.01$, γ = scale)	0	0	0.5563	0.5460
Poly(degree = 3, $C = 0.001$, γ = scale)	0	0	0.5300	0.5355
Sigmoid(γ = scale, $C = 1$)	0	0	0.5150	0.5340
Poly(degree = 4, $C = 0.01$, γ = scale)	0	0	0.5337	0.5230
Sigmoid(γ = scale, $C = 0.1$)	0	0	0.5356	0.5200
Sigmoid(γ = scale, $C = 10$)	0	0	0.5304	0.5140
Sigmoid(γ = scale, $C = 100$)	0	0	0.5303	0.5135
Poly(degree = 2, $C = 0.001$, γ = scale)	0	0	0.5135	0.5115

TABLE VI: Performance comparison of all SVM models for binary classification (cat vs dog)
Note: γ set to 'scale' is equal to 0.00033753205207176507 given that after PCA we have 103 features.

Appendix B
Multiclass Classification Figures and Tables

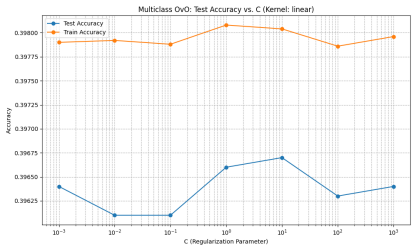


Fig. 23: Multiclass Classification (OvO): Linear Kernel - C vs Accuracy

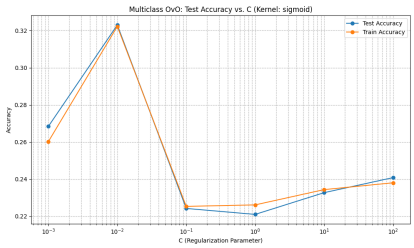


Fig. 24: Multiclass Classification (OvO): Sigmoid Kernel - C vs Accuracy

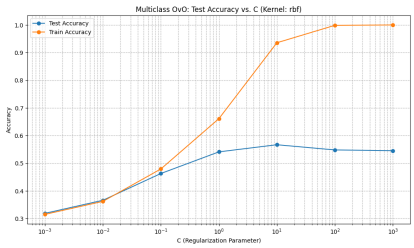


Fig. 25: Multiclass Classification (OvO): RBF Kernel - C vs Accuracy

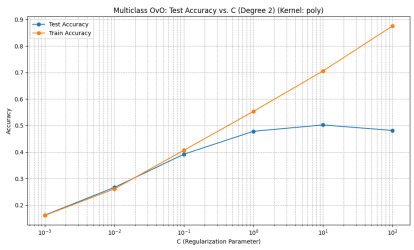


Fig. 26: Multiclass Classification (OvO): Polynomial Kernel (degree=2) - C vs Accuracy

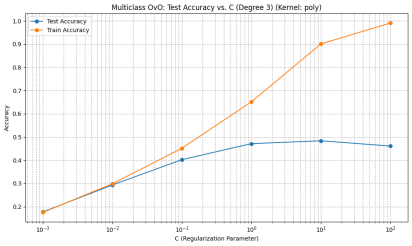


Fig. 27: Multiclass Classification (OvO): Polynomial Kernel (degree=3) - C vs Accuracy

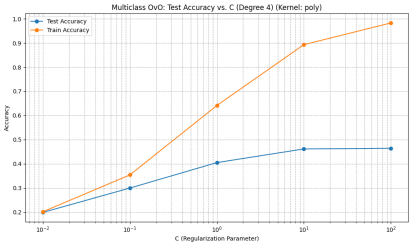


Fig. 28: Multiclass Classification (OvO): Polynomial Kernel (degree=4) - C vs Accuracy

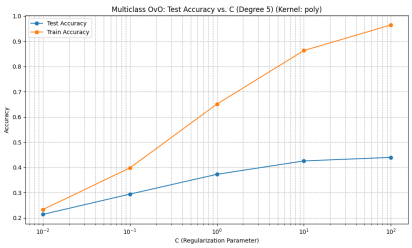


Fig. 29: Multiclass Classification (OvO): Polynomial Kernel (degree=5) - C vs Accuracy

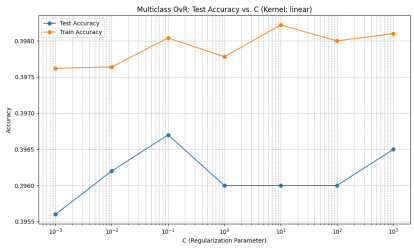


Fig. 30: Multiclass Classification (OvR): Linear Kernel - C vs Accuracy

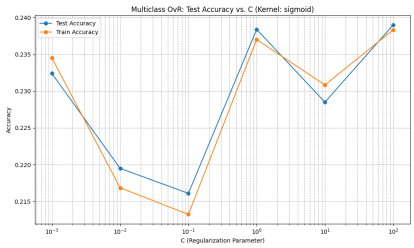


Fig. 31: Multiclass Classification (OvR): Sigmoid Kernel - C vs Accuracy

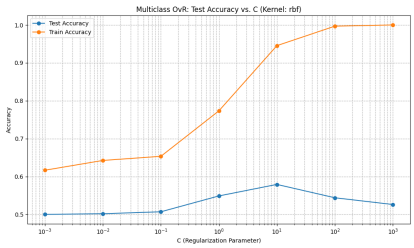


Fig. 32: Multiclass Classification (OvR): RBF Kernel - C vs Accuracy

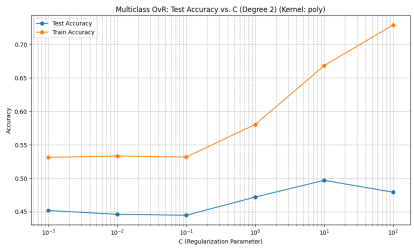


Fig. 33: Multiclass Classification (OvR): Polynomial Kernel (degree=2) - C vs Accuracy

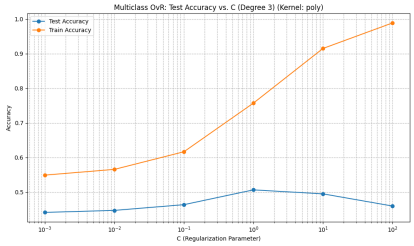


Fig. 34: Multiclass Classification (OvR): Polynomial Kernel (degree=3) - C vs Accuracy

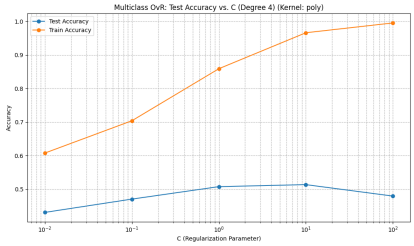


Fig. 35: Multiclass Classification (OvR): Polynomial Kernel (degree=4) - C vs Accuracy

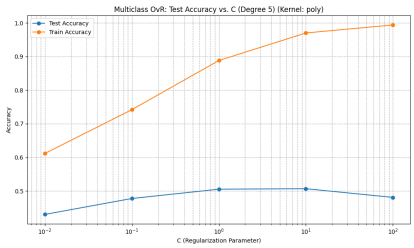


Fig. 36: Multiclass Classification (OvR): Polynomial Kernel (degree=5) - C vs Accuracy

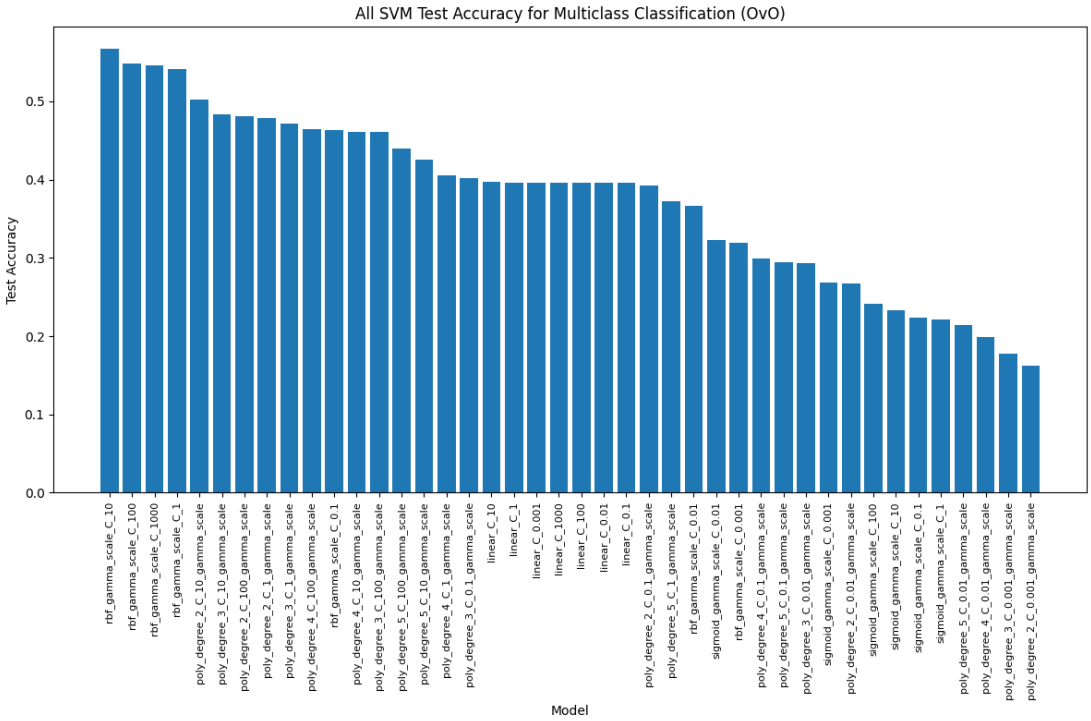


Fig. 37: Multiclass Classification (OvO): All SVM Models Test Accuracy Comparison

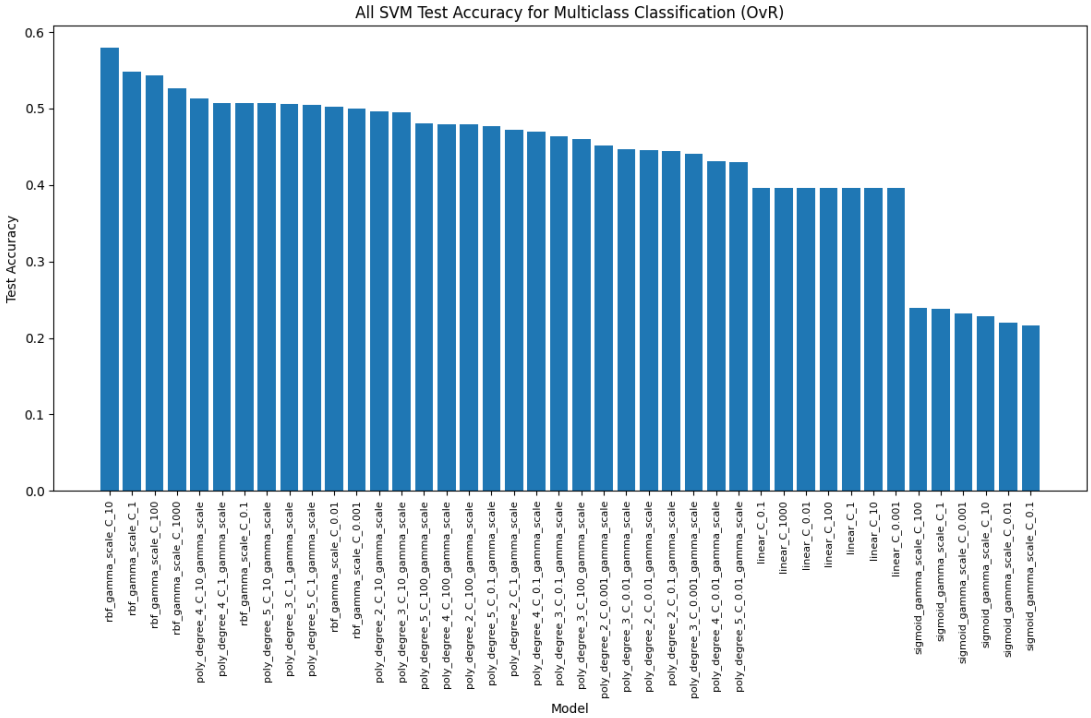


Fig. 38: Multiclass Classification (OvR): All SVM Models Test Accuracy Comparison

Model	Min	Sec	Train Acc.	Test Acc.
RBF(γ = scale, $C = 10$)	0	20	0.94534	0.5792
RBF(γ = scale, $C = 1$)	0	5	0.77330	0.5488
RBF(γ = scale, $C = 100$)	0	45	0.99684	0.5440
RBF(γ = scale, $C = 1000$)	0	55	1.00000	0.5262
Poly(deg=4, $C = 10$, γ = scale)	1	16	0.96590	0.5135
Poly(deg=4, $C = 1$, γ = scale)	0	28	0.85882	0.5075
RBF(γ = scale, $C = 0.1$)	0	3	0.65338	0.5070
Poly(deg=5, $C = 10$, γ = scale)	4	45	0.96994	0.5067
Poly(deg=3, $C = 1$, γ = scale)	0	26	0.75720	0.5064
Poly(deg=5, $C = 1$, γ = scale)	1	23	0.88800	0.5053
RBF(γ = scale, $C = 0.01$)	0	2	0.64248	0.5019
RBF(γ = scale, $C = 0.001$)	0	1	0.61666	0.5001
Poly(deg=2, $C = 10$, γ = scale)	0	53	0.66822	0.4969
Poly(deg=3, $C = 10$, γ = scale)	1	59	0.91502	0.4949
Poly(deg=5, $C = 100$, γ = scale)	10	28	0.99378	0.4809
Poly(deg=4, $C = 100$, γ = scale)	2	19	0.99508	0.4794
Poly(deg=2, $C = 100$, γ = scale)	6	53	0.72892	0.4792
Poly(deg=5, $C = 0.1$, γ = scale)	0	25	0.74194	0.4776
Poly(deg=2, $C = 1$, γ = scale)	0	16	0.58028	0.4719
Poly(deg=4, $C = 0.1$, γ = scale)	0	11	0.70380	0.4704
Poly(deg=3, $C = 0.1$, γ = scale)	0	12	0.61654	0.4636
Poly(deg=3, $C = 100$, γ = scale)	6	47	0.98878	0.4597
Poly(deg=2, $C = 0.001$, γ = scale)	0	4	0.53122	0.4518
Poly(deg=3, $C = 0.01$, γ = scale)	0	8	0.56560	0.4471
Poly(deg=2, $C = 0.01$, γ = scale)	0	6	0.53320	0.4461
Poly(deg=2, $C = 0.1$, γ = scale)	0	9	0.53172	0.4447
Poly(deg=3, $C = 0.001$, γ = scale)	0	5	0.54906	0.4413
Poly(deg=4, $C = 0.01$, γ = scale)	0	7	0.60760	0.4308
Poly(deg=5, $C = 0.01$, γ = scale)	0	11	0.61154	0.4303
Linear($C = 0.1$)	0	0	0.39804	0.3967
Linear($C = 1000$)	0	0	0.39810	0.3965
Linear($C = 0.01$)	0	0	0.39764	0.3962
Linear($C = 100$)	0	0	0.39800	0.3960
Linear($C = 10$)	0	0	0.39822	0.3960
Linear($C = 1$)	0	0	0.39778	0.3960
Linear($C = 0.001$)	0	0	0.39762	0.3956
Sigmoid(γ = scale, $C = 100$)	1	50	0.23834	0.2390
Sigmoid(γ = scale, $C = 1$)	0	1	0.23702	0.2384
Sigmoid(γ = scale, $C = 0.001$)	0	1	0.23452	0.2324
Sigmoid(γ = scale, $C = 10$)	0	1	0.23084	0.2285
Sigmoid(γ = scale, $C = 0.01$)	0	1	0.21686	0.2195
Sigmoid(γ = scale, $C = 0.1$)	0	1	0.21324	0.2161

TABLE VIII: Multiclass One-vs-Rest SVM performance comparison

Model	Min	Sec	Train Acc.	Test Acc.
RBF(γ = scale, $C = 10$)	0	9	0.93558	0.5669
RBF(γ = scale, $C = 100$)	0	15	0.99860	0.5481
RBF(γ = scale, $C = 1000$)	0	16	1.00000	0.5455
RBF(γ = scale, $C = 1$)	0	2	0.66114	0.5411
Poly($d = 2$, $C = 10$, γ = scale)	0	20	0.70580	0.5024
Poly($d = 3$, $C = 10$, γ = scale)	0	19	0.90086	0.4837
Poly($d = 2$, $C = 100$, γ = scale)	2	4	0.87550	0.4816
Poly($d = 2$, $C = 1$, γ = scale)	0	4	0.55352	0.4784
Poly($d = 3$, $C = 1$, γ = scale)	0	6	0.65164	0.4714
Poly($d = 4$, $C = 100$, γ = scale)	0	18	0.98288	0.4643
RBF(γ = scale, $C = 0.1$)	0	2	0.47936	0.4629
Poly($d = 4$, $C = 10$, γ = scale)	0	12	0.89314	0.4615
Poly($d = 3$, $C = 100$, γ = scale)	0	41	0.99106	0.4610
Poly($d = 5$, $C = 100$, γ = scale)	0	26	0.96482	0.4400
Poly($d = 5$, $C = 10$, γ = scale)	0	14	0.86342	0.4260
Poly($d = 4$, $C = 1$, γ = scale)	0	6	0.64132	0.4050
Poly($d = 3$, $C = 0.1$, γ = scale)	0	4	0.45166	0.4025
Linear($C = 10$)	0	0	0.39804	0.3967
Linear($C = 1$)	0	0	0.39808	0.3966
Linear($C = 1000$)	0	0	0.39796	0.3964
Linear($C = 0.001$)	0	0	0.39790	0.3964
Linear($C = 100$)	0	0	0.39786	0.3963
Linear($C = 0.01$)	0	0	0.39792	0.3961
Linear($C = 0.1$)	0	0	0.39788	0.3961
Poly($d = 2$, $C = 0.1$, γ = scale)	0	4	0.40708	0.3921
Poly($d = 5$, $C = 1$, γ = scale)	0	7	0.65114	0.3729
RBF(γ = scale, $C = 0.01$)	0	2	0.36202	0.3661
Sigmoid(γ = scale, $C = 0.01$)	0	2	0.32220	0.3230
RBF(γ = scale, $C = 0.001$)	0	2	0.31502	0.3188
Poly($d = 4$, $C = 0.1$, γ = scale)	0	4	0.35408	0.2997
Poly($d = 5$, $C = 0.1$, γ = scale)	0	5	0.39840	0.2945
Poly($d = 3$, $C = 0.01$, γ = scale)	0	4	0.29856	0.2935
Sigmoid(γ = scale, $C = 0.001$)	0	2	0.26022	0.2684
Poly($d = 2$, $C = 0.01$, γ = scale)	0	4	0.26116	0.2670
Sigmoid(γ = scale, $C = 100$)	0	1	0.23802	0.2408
Sigmoid(γ = scale, $C = 10$)	0	1	0.23434	0.2327
Sigmoid(γ = scale, $C = 0.1$)	0	1	0.22532	0.2242
Sigmoid(γ = scale, $C = 1$)	0	1	0.22612	0.2210
Poly($d = 5$, $C = 0.01$, γ = scale)	0	4	0.23336	0.2141
Poly($d = 4$, $C = 0.01$, γ = scale)	0	4	0.20172	0.1989
Poly($d = 3$, $C = 0.001$, γ = scale)	0	4	0.17536	0.1775
Poly($d = 2$, $C = 0.001$, γ = scale)	0	4	0.16142	0.1625

TABLE VII: Multiclass One-vs-One SVM performance comparison

Appendix C
Confusion Matrices of Best Models

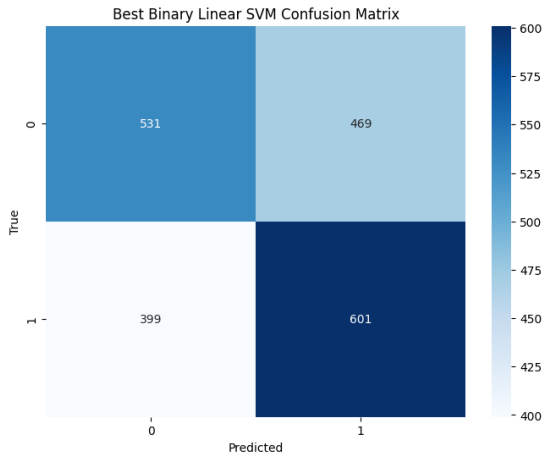


Fig. 39: Binary Classification: Confusion Matrix of the best linear SVM model

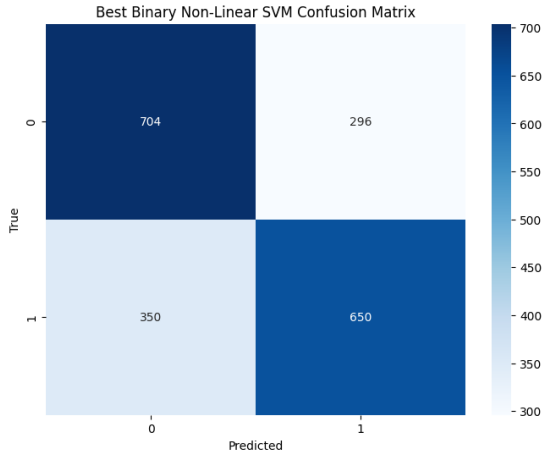


Fig. 40: Binary Classification: Confusion Matrix of the best nonlinear (RBF) SVM model

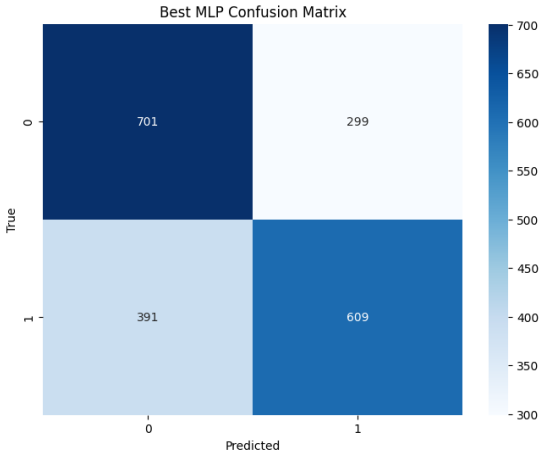


Fig. 41: Binary Classification: Confusion Matrix of the best MLP model

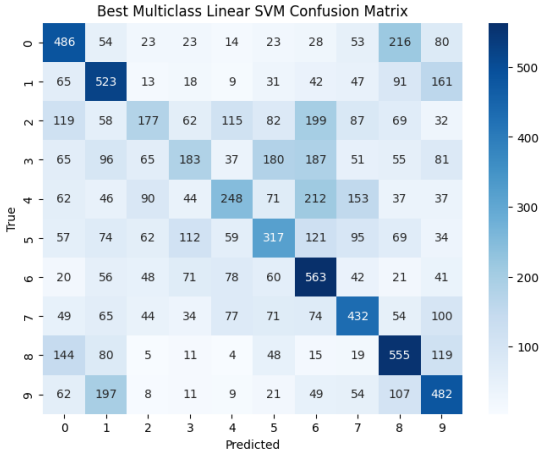


Fig. 42: Multiclass Classification: Confusion Matrix of the best linear SVM model

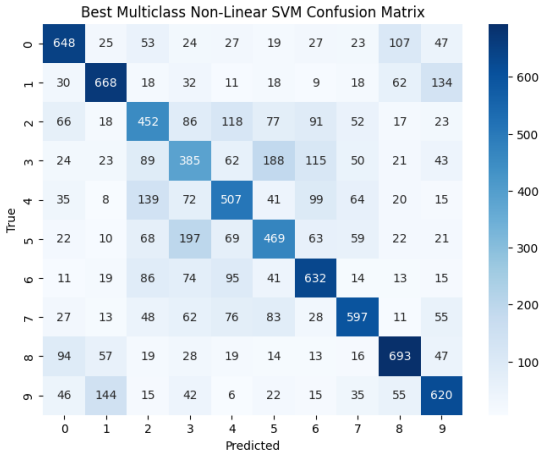


Fig. 43: Multiclass Classification: Confusion Matrix of the best nonlinear (RBF) SVM model

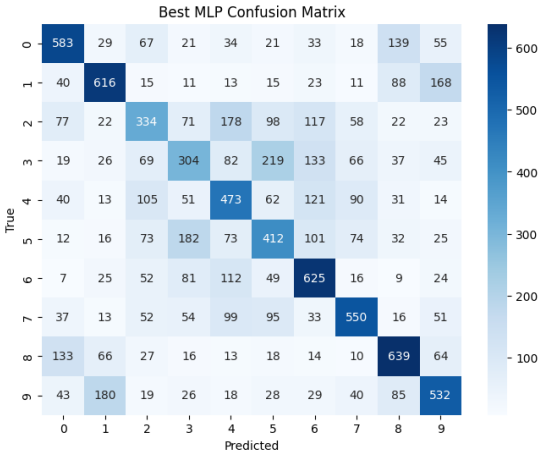


Fig. 44: Multiclass Classification: Confusion Matrix of the best MLP model